SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

Annexure A

Dry Run Evaluation

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:



Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

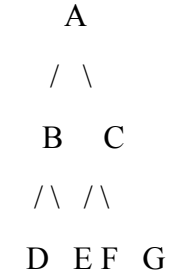
Iteration Queue Visited Node

1
2
3
4
5
6
7

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,



Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**
   1 3 6
   5 0 2
   4 7 8

Goal State :
   1 2 3
   4 5 6
   7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration Current State Queue Before Expansion Queue After Expansion
1
2
3
4
5

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

QUESTION 1 – BREADTH-FIRST SEARCH (BFS)

Given graph: A has children B and C; B has children D and E; C has children F and G. Assume children are visited from left to right.

1. BFS Traversal Order

BFS visits nodes level by level using a FIFO (First-In, First-Out) queue.

Traversal: A → B → C → D → E → F → G

2. Queue Contents After Each Iteration

The queue below is shown after removing/expanding the current node and adding its unvisited children.

Iteration	Expanded / Visited Node	Queue After Expansion
1	A	B, C
2	B	C, D, E
3	C	D, E, F, G
4	D	E, F, G
5	E	F, G
6	F	G
7	G	Empty

Initial queue: [A]. Visited order: A, B, C, D, E, F, G.

BFS Procedure

1. Insert the start node A into the queue and mark it discovered.
2. Remove the front node from the queue and visit it.
3. Insert its undiscovered children at the rear of the queue.
4. Repeat until the queue is empty or the required goal is found.

QUESTION 2 – DEPTH-FIRST SEARCH (DFS)

Use the same graph and visit children from left to right. DFS uses a stack or recursion and explores one branch as deeply as possible before backtracking.

DFS traversal: A → B → D → E → C → F → G

Step	Node Visited	Action / Explanation
1	A	Start at A; go to left child B.
2	B	Go to left child D.
3	D	Leaf node; backtrack to B.
4	E	Visit B's remaining child; then backtrack to A.
5	C	Visit A's remaining child; go to

		F.
6	F	Leaf node; backtrack to C.
7	G	Visit C's remaining child; traversal ends.

BFS vs. DFS

Feature	BFS	DFS
Data structure	FIFO queue	Stack / recursion
Exploration	Level by level	Deep branch first
This graph's order	A, B, C, D, E, F, G	A, B, D, E, C, F, G
Shortest path	Finds shortest path in unweighted graphs	Not guaranteed

QUESTION 3 – 8-PUZZLE USING BFS

Initial and Goal States

Initial state:

1 3 6\n5 0 2\n4 7 8

Goal state:

1 2 3\n4 5 6\n7 8 0

The blank (0) can move up, down, left, or right when the move remains inside the board. BFS explores states by increasing move depth.

1. Complete Shortest Solution Path

State 0 – Start (Initial)

1 3 6\n5 0 2\n4 7 8

State 1 – Right

1 3 6\n5 2 0\n4 7 8

State 2 – Up

1 3 0\n5 2 6\n4 7 8

State 3 – Up

1 0 3\n5 2 6\n4 7 8

State 4 – Right

1 2 3\n5 0 6\n4 7 8

State 5 – Down

1 2 3\n0 5 6\n4 7 8

State 6 – Right

1 2 3\n4 5 6\n0 7 8

State 7 – Right

1 2 3\n4 5 6\n7 0 8

State 8 – Down (Goal)

1 2 3\n4 5 6\n7 8 0

Solution moves for the blank: Right → Up → Up → Right → Down → Right → Right → Down.

Minimum solution length: 8 moves.

2. Which Node Is Expanded in Each Iteration?

BFS expands states in FIFO order, by nondecreasing depth. The path above shows the solution states, but these are not necessarily the first states expanded: other states at the same depths are examined first according to the successor order.

For the stated move ordering (Up, Down, Left, Right, omitting illegal moves), BFS reaches the goal after expanding 283 states; 456 distinct states have been discovered, including frontier states. Counts depend on successor ordering and on whether "generated" counts repeated states or only new unique states.

3. How Many States Are Generated in Total?

With duplicate-state checking and successor order Up, Down, Left, Right, the search discovers 456 unique states by the time the goal is dequeued. Of these, 283 states have been expanded. These are the dry-run totals under the stated convention; counting every attempted successor including duplicates would give a different number.

4. Complete Solution Path

Initial → State 1 → State 2 → State 3 → State 4 → State 5 → State 6 → State 7 → Goal (states listed above).

5. Why BFS Finds the Shortest Solution

Every legal move in the standard 8-puzzle has equal cost (one move). BFS explores all states at depth 0, then depth 1, then depth 2, and so on. Therefore, the first time it reaches the goal, no shallower solution can exist. BFS is complete for this finite state space, assuming repeated states are handled and the goal is reachable.